KAUST Thuwal Campus Finals, 3-Seat Relay & Academy Synthesis

KAUST Thuwal Campus Finals, 3-Seat Relay & Academy Synthesis Practice Exercises

1 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/5}$ be a primitive 5-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{4} (2 - \zeta^k)$.

a. 32

b. 17

c. 31

d. 27

2 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/6}$ be a primitive 6-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{5} (2 - \zeta^k)$.

a. 64

b. 63

c. 33

d. 59

3 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/7}$ be a primitive 7-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{6} (2 - \zeta^k)$.

a. 127

b. 128

c. 65

d. 123

4 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/8}$ be a primitive 8-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{7} (2 - \zeta^k)$.

a. 256

b. 129

c. 251

d. 255

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5 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 6 \\ j \equiv 0 \pmod 3}} \binom{6}{j}$.

a. 21

b. 24

c. 22

d. 16

6 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 6 \\ j \equiv 1 \pmod 3}} \binom{6}{j}$.

a. 23

b. 21

c. 16

d. 22

7 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 7 \\ j \equiv 2 \pmod 3}} \binom{7}{j}$.

a. 42

b. 44

c. 32

d. 43

8 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 7 \\ j \equiv 0 \pmod 3}} \binom{7}{j}$.

a. 42

b. 45

c. 32

d. 43

9 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{10} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^2)(1-x^5)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 2v + 5w = 10$.

a. 12

b. 8

c. 10

d. 14

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10 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{11} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^2)(1-x^3)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 2v + 3w = 11$.

a. 18

b. 16

c. 14

d. 20

11 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{12} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^3)(1-x^5)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 3v + 5w = 12$.

a. 9

b. 11

c. 7

d. 13

12 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{13} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^2)(1-x^4)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 2v + 4w = 13$.

a. 18

b. 14

c. 20

d. 16

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